SUPERCONDUCTIVITY

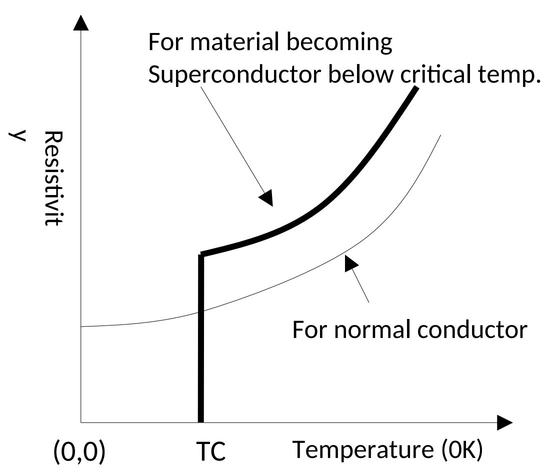
Q1. Define superconductivity.

(M.U. Dec 2007, 2008, 2009, 2011, 2012, 2016; May 2008, 2010, 2012, 2013, 2014, 2015, 2016) [2 Marks]

Superconductivity is a phenomenon of sudden disappearance of electrical resistance and expulsion of magnetic flux occurring in certain materials when they are cooled below a characteristic low temperature.

Q2. Define Critical temperature.

(M.U. May 2008, 2010, 2012, 2013, 216, 2017; Dec. 2007, 2010, 2016) [2 Marks]



When superconducting material is cooled below a certain temperature, its resistance suddenly drops to zero and it goes into the superconducting state from the normal state. The temperature at which a material transforms into superconducting state is called critical temperature 'T_c' for that material.

Different materials have different critical temperatures. The transition is reversible. When the temperature of the material is increased above the critical temperature, it passes into the normal state. For elementary solids (in extremely pure form), critical temperature (T_c) is found to be very low (e.g. Tungsten → 0.015° K, Zinc → 0.85°K, Niobium → 9.46°K), whereas for compounds or alloys, T_c is found relatively high. {Nb Ti →10^0K, Nb_3Ge→23.2,^{0}K, etc.}

Critical Temperature is the temperature below which the electrical resistance of a material suddenly drops to zero is called critical temperature of that material.

Q3. Define Critical Field.

(M.U. Dec. 2010, 2016) [1 Mark]

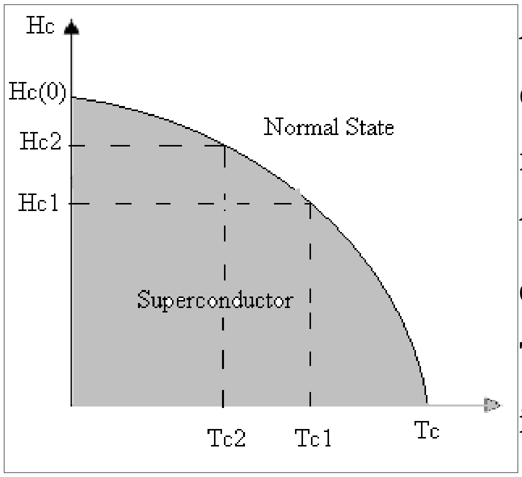


Fig. 4.2 Change in the critical field of a material while it is subjected to change in temperature

A material in its superconducting state, behaves like a diamagnetic material when placed in a weak magnetic field. But if the field strength is increased, the material may lose superconductivity even below critical temperature (T_c).

The critical field (H_c) for a superconducting material is the minimum field value at which normal resistivity is regained by the material and loses its superconducting state. Naturally, H_c value depends on temperature of superconductor which is placed in external magnetic field as shown in Fig.4.2. At any temperature less than T_c, the superconducting material when placed in external magnetic field, remains superconductor below H_c (value at that temperature) and is normal above the H_c.

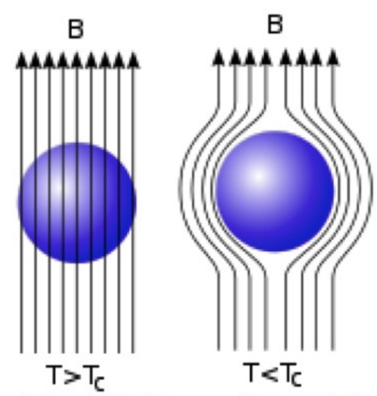
$$H_c(T)=H_c(0)\left[1-\left(\frac{T}{T_c}\right)^2\right]$$

For pure mercury, $H_c = 0.04$ Tesla --- at $T \approx 0^0$K

$H_c = 0.02$ Tesla --- at $T \approx 3^0$K

Q4. Explain Meissner Effect with the help of diagrams.

(M.U. May 2008, 2012; Dec. 2012, 2018; June 2019) [5 Marks]



Superconductor at $T>T_c$ in weak external magnetic field

Superconductor at $T<T_c$ in weak external magnetic field

When a superconducting material is cooled in a weak magnetic field (H < Hc), at a certain low temperature called the critical temperature (Tc), the field flux is expelled from the body of the superconducting material, which has just entered superconducting state. This phenomenon was discovered by Meissner and Oschenfield in 1933 and called as Meissner's effect (Flux expulsion).

When a superconductor is in normal state the external magnetic field lines are able to penetrate through its body but when it is in a superconducting state the field lines (i.e. magnetic flux) are strongly expelled from the body. This observation explains why a superconductor in a superconducting state is said to be perfect diamagnetic material.

The magnetic induction inside a material is given by

$$B=\mu_0 (H+M)$$

Where H denotes the external field strength and M denotes the magnetization produced within specimen.

For superconductor to be in superconducting state, i.e. when $T < T_C$

B (total magnetic induction inside specimen) = 0

- ⇨ $H+M=0 \quad \Rightarrow \quad M=-H$

Which means magnetic susceptibility is negative and this is the perfect criterion for diamagnetic substance

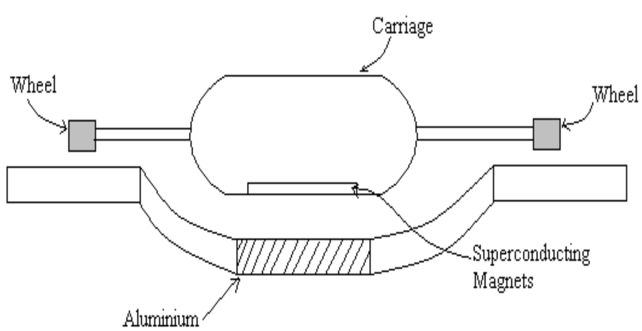
Magnetic susceptibility, $\chi=\frac{M}{H}=-1$

Thus, magnetic flux is excluded by superconducting material, because an equal and opposite magnetization is induced inside its body.

Because of this effect, a superconductor strongly repels an external magnet (smaller in size). Hence such a small magnet hovers over a superconducting body leading to 'levitation effect' or in other way, if a small sized superconductor is taken with a big magnet, it will hang beneath the magnet showing 'suspension effect'.

Thus, Meissner's effect is useful in

1. Confirming superconductivity: When a material being cooled in weak magnetic field (H < Hc) demonstrates flux expulsion, we can confirm that the material has a superconducting behavior. This is because suddenly the magnetic field lines are pushed out of the body which were initially penetrating through body.
2. Levitated frictionless bearings: The machines with rotating shafts ball bearings are covered with lubricants to reduce friction and thereby increase rotations per minute (rpm). With a magnetic shaft and superconducting bearings (or vice versa) the levitation effect will not allow any friction among the surfaces and a high rotation speed can be achieved without any power loss against friction.
3. Levitating trains (MAGLEV): MAGLEV stands for Magnetically Levitated vehicles.



The coaches of the train do not slide over the steel rails but float on a four-inch air cushion above the track using superconducting magnets; this eliminates friction and energy loss as heat, allowing the train to reach high speeds of the order of 500 km/hr. Such magnetic levitation trains would make train travel much faster, smoother, and more efficient due to lack of friction between the tracks and train.

Working: A typical plan of MAGLEV train is shown in Fig.4.4. The train has superconducting magnets built into the base of carriages. An aluminum guideway is laid on the ground and carries electric current. The repulsion between the field produced by the superconductor and the field produced by electric current in the aluminum guideway causes magnetic levitation of the train. The train is fitted with retractable wheels. Once the train is levitated in the air, the wheels are retracted into the body and train glides forward on the air cushion.

When the train is to be halted, the wheels are drawn out and the train descends slowly onto the guideway and runs forward till to stop.

Q5. What is Type-I and Type-II Superconductors.

(M.U. May 2014, 2015; Dec. 2008 2009, 2011, 2012) [5 Marks]

On the basis of their magnetic behavior, superconductors are classified as Type-I and Type-II.

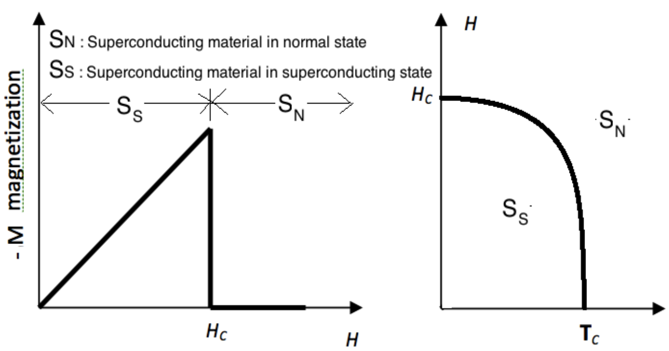
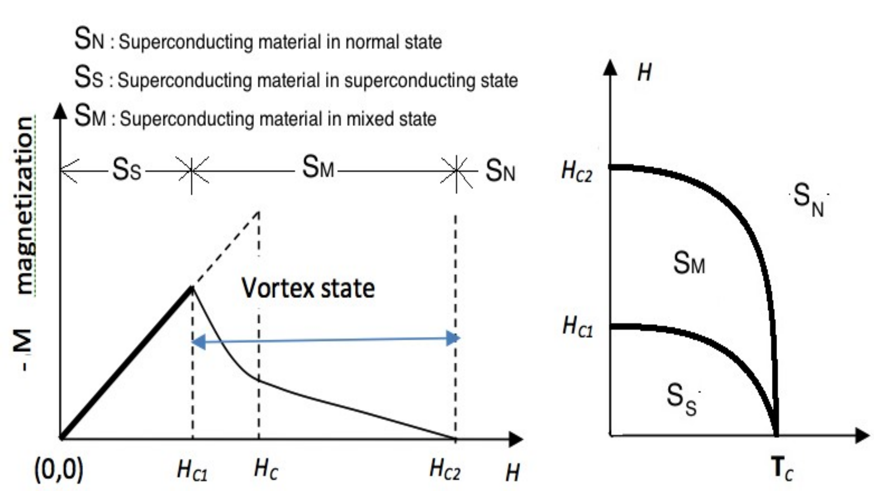


Fig. 4.5 (a) Change in the diamagnetic behavior of the Type-I superconductor with respect to change in the applied magnetic field (H); (b) The change in the state of superconductor from normal (S_n) to superconducting (S_s) with respect to the applied magnetic field and temperature

Type I superconductors demonstrate a perfect diamagnetic behavior when in their superconducting state (SS). They get magnetized in the direction opposite to that of the applied field and their magnetization increasing with the increasing magnetic field. When the applied field is increased to the critical field (H_c) the type-I superconductor loses superconductivity and also its diamagnetic property thereby entering a normal state (SN). Due to this the magnetization abruptly drops down to zero and material returns to normal state as shown in Figure 4.3(a). Figure 4.3(b) shows the plot of magnetic field verses temperature indicating the region where the superconducting material exists in a superconducting state (SS) and the region above critical field (HC) and critical temperature (TC) where it enters a normal state (SN).

Type I superconductors generally have very low value of critical field (HC ~ 0.05 wb/m^2). Due to this reason Type I superconductors very susceptible to stray magnetic field and hence their applications are limited.



Type II superconductivity was discovered in 1930. In such superconductors the transition from superconducting state to normal state occurs gradually. Type-II superconductors are characterized by two critical field value H_{C1} and H_{C2}.

As shown in Figure 4.4 (a) the magnetization of Type-II superconductor increases with the increase in applied magnetic field till the lower critical field H_{C1}. After this the magnetization decreases gradually with increasing applied field till it reaches zero at field value H_{C2}.

The material is in its superconducting state below H_{C1}, after H_{C2} the superconductor enters its normal state, whereas between H_{C1} and H_{C2} the superconducting material exists in a mixed state as shown in Fig. 4.4 (b).

A superconducting state doesn't allow magnetic field to penetrate inside the body. When H_{C1} is crossed, few magnetic field lines 'leak' through the material, which means the thread like region of material through which the field passes get back to normal state whereas remaining body of the material is in the superconducting state. Thus, after H_{C1}, the superconducting material has threadlike 'drills' of normal state in it due to the magnetic field hence the mixed state is also called as the vortex state. When applied field reaches value H_{C2} these 'normal state drills' increase in number and occupy the whole body of the superconducting material transforming it completely into the normal state.

Type-II Superconductor, provide a liberty of enjoying a partially superconducting state up to H_{C2}, which is relatively larger ($H_{C2} \sim 20$ wb/m^2). Hence, Type-II superconductors are widely used industries.

Q6. What is SQUID.

A superconducting quantum interference device (SQUID) is a mechanism used to measure extremely weak magnetic flux, such as small changes in the human body's electromagnetic energy field using a Josephson junction. The beauty of the device is that it can detect a change of energy as small as 100 billion times weaker than the electromagnetic energy that moves a compass needle.

(or can detect extremely small changes in magnetic flux (about 10^{-14} T....10^{-15} T)).

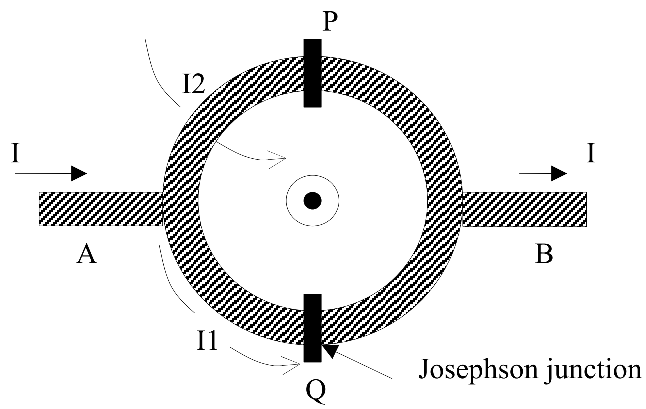


Fig. 4.9Diagrammatic representation of SQUID

The heart of SQUID is superconducting ring, which contains one or more Josephson junctions. A SQUID consists of two Josephson junctions in parallel and relies on the interference of the current from each function.

It consists of a ring of superconducting material having two side arms A and B which acts as entrance and exit for the supercurrent respectively. The insulating layers P and Q may in general, have the different thickness and let the current through these layers be I_1 and I_2 respectively.

A highly coherent supercurrent coming along the path I is divided along two paths. The superconducting state is represented by a wave function which experience a phase shift at the junctions P and Q.

Let the phase difference between points A and B taken on a path through junction P be δP. When taken on a path through Q the phase difference is δQ. In the absence of magnetic field these two phases are equal.

$\delta P - \delta Q = 0$ in the absence of an external magnetic field.

When a magnetic field is applied, magnetic flux is enclosed by a closed circuit, the phase difference around the closed circuit is given, according to theory, by

$$\delta P - \delta Q = \frac{2e}{\hbar c}\varphi$$

In supercurrent interference, the waves are matter waves of cooper pairs and the phase difference is caused by the applied magnetic field.

Both I_1 and I_2 vary periodically with the total flux through the area of SQUID. The current is sensitive to very small changes in the magnetic field. So, the SQUID can be used as a very sensitive galvanometer.

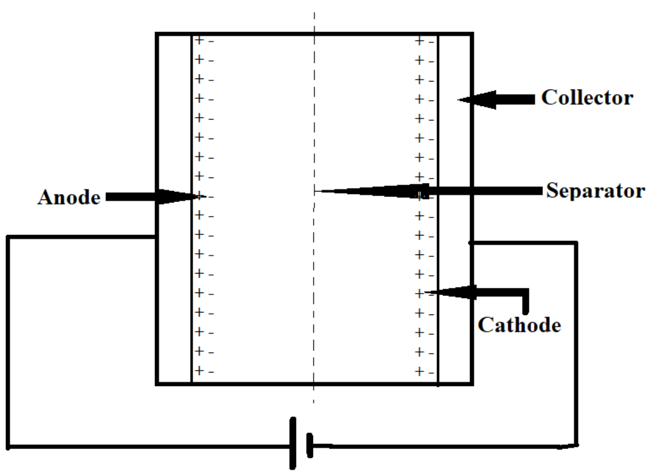
SQUIDs are used in medical diagnostic equipment's to detect minute changes in magnetic fields generated by human brain or body.

Q7. Explain Supercapacitor. **(M.U. Dec. 2019) [5 Marks]**

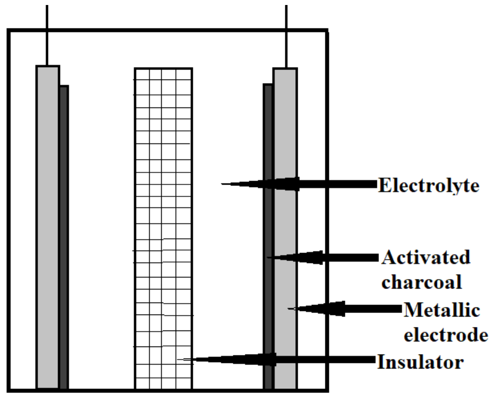
It is a capacitor which has high capacitance and capable of changing and storing energy at a higher density than normal capacitor.

Principle:

Supercapacitor works on the principle of double layer capacitance at the electrode - electrolyte interface where electric charges are accumulated on the electrode surface and ions of opposite charge are arranged on the electrolyte side.



Construction:



In supercapacitors, plates are made of metals coated with a porous substance. Porous material gives larger surface area increasing the storage capacity.
These plates are soaked in an electrolyte and are separated by a thin insulator.
Post charging of this supercapacitor, layers of opposite charges are formed on both the sides of the separator, thus forming the double layer.

Types of Supercapacitor

1. Double Layer Capacitors

These capacitors achieve energy storage with activated carbon electrodes.

2. Pseudo-capacitors

These capacitors store electrical energy by electron charge transfer between electrodes and electrolyte.
Metal oxide or conductive polymer electrodes are used.

3. Hybrid Supercapacitor

These have two electrodes; one stores charge electrostatically while other exhibits electrochemical capacitance (farcically). Thus, it's a combination of Double layer capacitor and Electrochemical pseudo capacitor.